IndicSentEval: How Effectively do Multilingual Transformer Models encode Linguistic Properties for Indic Languages?

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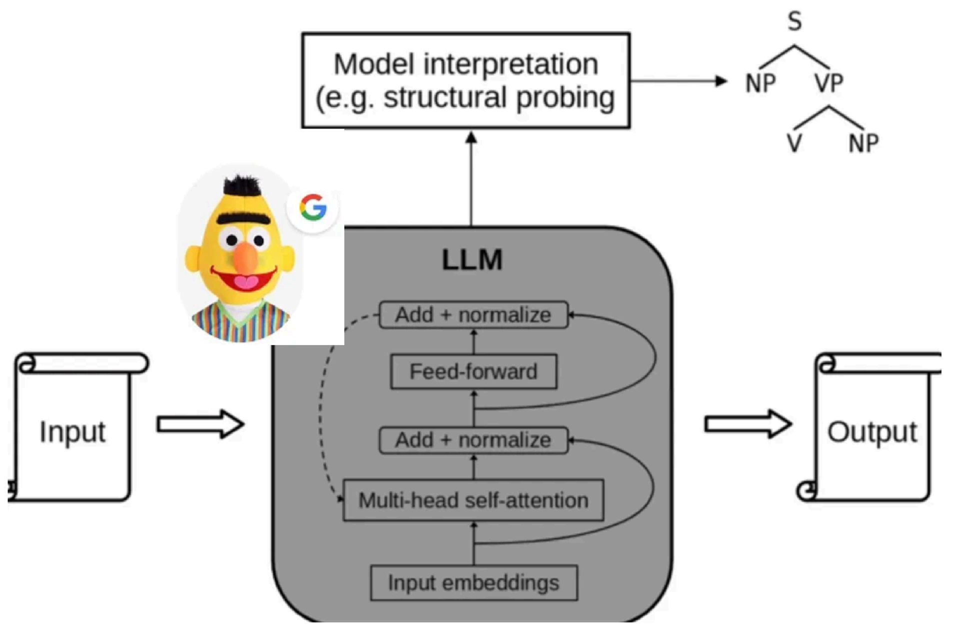
Introduction

BERTology studies focused on investigating the internal workings and linguistic representations of language models (LM) [1,2, 3]



Investigate via a probing tasks

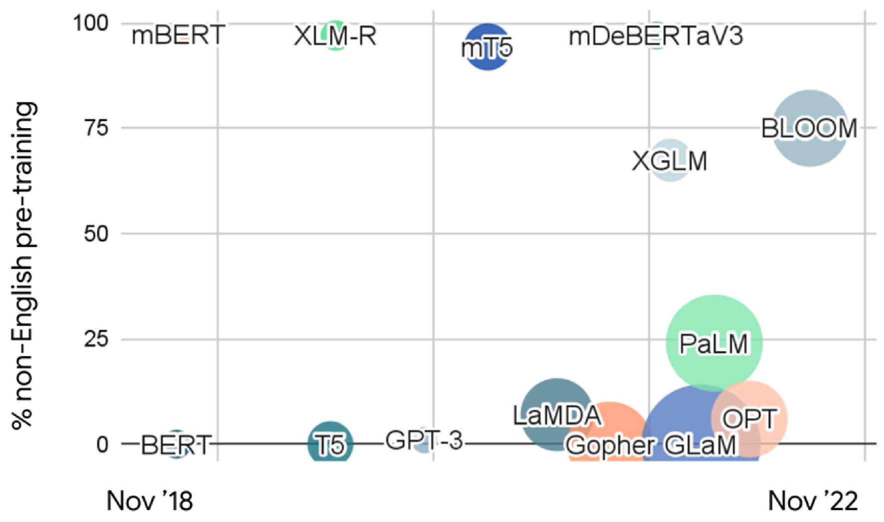
Tasks:

- **Suface** – Sentence Length, Word Content
- **Syntactic** – Bigram shift, Tree depth, Top constituent
- **Semantic** – Tense, Subject Number, Object Number

Hierarchy of linguistic info ⇒ how BERT encodes linguistic properties across layers

Multi-lingual language models are pretrained on many languages and learn representations for each language

Universal multi-lingual models



Pretrained on 100+ languages

Indic multi-lingual models

Pretrained on 11+ languages

"How multi-lingual language models capture linguistic properties across layers for different Indic languages?"

Methods

IndicSentEval: multi-lingual probing tasks for Indic languages

Languages: Hindi, Marathi, Urdu, Telugu, Kannada, Malayalam

Models: mBERT, XLM, InfoXLM, MuRIL, IndicBERT, BLOOM, mT5, mGPT, XGLM

IndicSentEval

Probing Tasks

- **Surface:** SentLen
- **Syntactic:** BShift, TreeDepth
- **Semantic:** SubjNum, ObjNum, VerbGen, VerbNum, VerbPer

Perturbations: AppendR, DropNV, DropN, DropV, DropF, DropFL, DropL, DropRN, DropNV, KeepNV, KeepN, KeepV, Shuffle

IndicSentEval Dataset

Task	|C|	Labels
SentLen	8	(0-5),(6-8),(9-12),(13-16),(17-20),(21-25),(26-28),(29-32)
TreeDepth	5	(0-2),(3-5),(6-8),(9-11),(12-20)
BShift	2	0,1
SubjNum	2	singular, plural
ObjNum	2	singular, plural
VerbGen	4	masculine, feminine, neutral, any
VerbNum	3	singular, plural, any
VerbPer	7	1^{st} person, 2^{nd} person, 3^{rd} person, 1^{st} person honorific, 2^{nd} person honorific, 3^{rd} person honorific, any

$$A_c^{(l)} = \mathrm{Eval}\Big(\mathrm{Probe}\big(f_{\mathrm{LM}}^{(l)}(x)\big)\Big)$$

$A_c^{(l)}$: probing accuracy at layer l

x: probing task input example

$f_{LM}^{(l)}(x)$: LM representation at layer l

Probe(·): classifier trained on probing labels

Eval(·): evaluation metric (e.g., accuracy)

Hindi	जोर से आवाज लगाने पर भी यह नजारा देखा जा सकता है । This scene can be seen even after making loud noise.	SentLen → 13; TreeDepth → 3; SubjNum → plural (pl); ObjNum → singular (sg); BShift → जोर से आवाज लगाने पर भी यह नजारा **जा देखा** सकता है । (This scene can **seen be** even after making loud noise) ; VerbGen → male (m); VerbNum → singular (sg); VerbPer → any (any)

IndicSentEval Text Perturbations

Text-Perturbation	Perturbed Sentence	Perturbed Sentence (Translation in English)
Original	डलहौजी उस धौलाधार पर्वतमाला के सामने पड़ता है जो साल भर बर्फ़ की नई नई परतें ओढ़ती है ।	Dalhousie lies in front of the Dhauladhar range which is covered with new layers of snow throughout the year.
AppendR	डलहौजी उस धौलाधार पर्वतमाला के सामने पड़ता है जो लपकने साल भर बर्फ़ की नई नई परतें ओढ़ती है ।	Dalhousie lies in front of the Dhauladhar range which is covered with new layers of snow throughout the year catch.
DropNV	उस के जो भर की नई नई ।	in of the which with new of throughout the.
DropN	उस के पड़ता है जो भर की नई नई ओढ़ती है ।	lies in of the which is covered with new of throughout the.
DropV	डलहौजी उस धौलाधार पर्वतमाला के सामने जो साल भर बर्फ़ की नई नई परतें ।	Dalhousie in front of the Dhauladhar range which with new layers of snow throughout the year.
DropF	उस धौलाधार पर्वतमाला के सामने पड़ता है जो साल भर बर्फ़ की नई नई परतें ओढ़ती है ।	lies in front of the Dhauladhar range which is covered with new layers of snow throughout the year.
DropFL	उस धौलाधार पर्वतमाला के सामने पड़ता है जो साल भर बर्फ़ की नई नई परतें ओढ़ती ।	lies in front of the Dhauladhar range which is covered with new layers of snow throughout the.
DropL	डलहौजी उस धौलाधार पर्वतमाला के सामने पड़ता है जो साल भर बर्फ़ की नई नई परतें ओढ़ती ।	Dalhousie lies in front of the Dhauladhar range which is covered with new layers of snow throughout the.
DropRN	डलहौजी उस धौलाधार पर्वतमाला के पड़ता है जो साल भर बर्फ़ की नई नई परतें ओढ़ती है ।	Dalhousie lies in front of the range which is covered with new layers of snow throughout the year.
DropRV	डलहौजी उस धौलाधार पर्वतमाला के सामने पड़ता जो साल भर बर्फ़ की नई नई परतें ओढ़ती है ।	Dalhousie lies in front of the Dhauladhar range which is with new layers of snow throughout the year.
KeepNV	डलहौजी धौलाधार पर्वतमाला सामने पड़ता है साल बर्फ़ परतें ओढ़ती है	Dalhousie lies front Dhauladhar range is covered layers snow year.
KeepN	डलहौजी धौलाधार पर्वतमाला सामने साल बर्फ़ परतें	Dalhousie front Dhauladhar range layers snow year.
KeepV	पड़ता है ओढ़ती है	lies is covered.
Shuffle	। के उस की है नई पड़ता जो भर डलहौजी नई बर्फ़ साल ओढ़ती परतें सामने पर्वतमाला धौलाधार है	.with which new lies Dalhousie snow year covered layers in front of the range Dhauladhar is.

$$RS = 1 - \frac{A_c - A_p}{A_c}$$

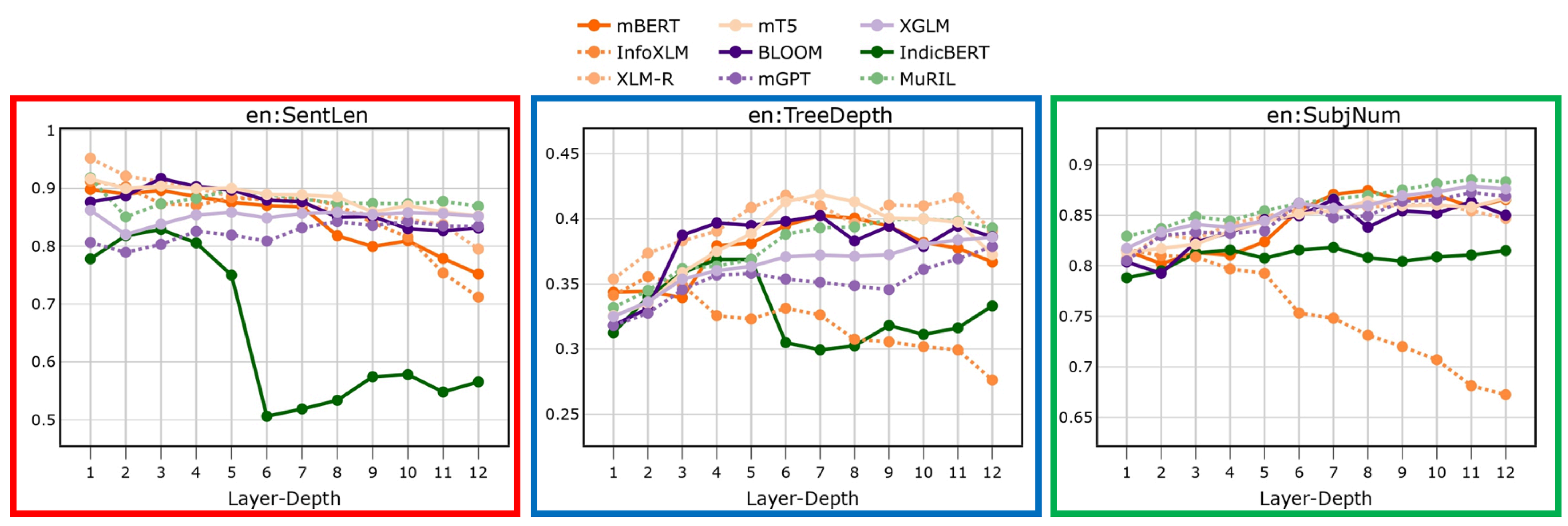
RS → Robustness Score
A_c → Original Accuracy
A_p → Perturbed Accuracy

References

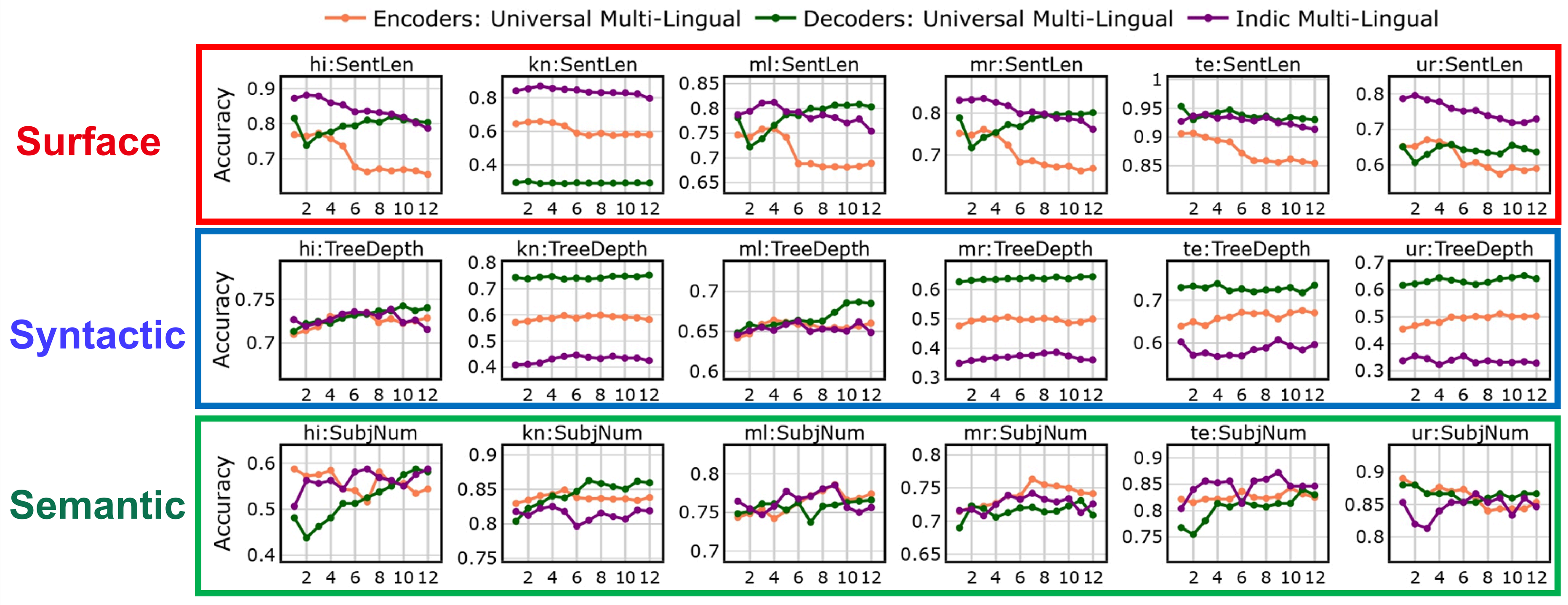
[1] Conneau et al. 2018. What you can cram into a single $ &!#* vector: Probing sentence embeddings for linguistic properties. ACL-2018
[2] Jawahar et al. 2019. What does bert learn about the structure of language? ACL-2019
[3] Rogers et al. 2020. A primer in bertology: What we know about how bert works. TACL-2020

Results

For En-> How multilingual models encode linguistic properties across layers



Probing Results: Universal vs. Indic multi-lingual language models



- **Indic-specific models:** best at capturing language properties within the realm of Indic languages, may be due to their targeted training.
- **Universal multi-lingual models:** Both encoder-based and decoder-based models show mixed results, may be due to their broader training across languages.

Perturbation Results: Universal vs. Indic multi-lingual language models

Multilingual models vs. Indic languages

	hi	kn	ml	mr	te	ur
mBERT	0.794	0.736	0.583	0.626	**0.980**	0.715
IndicBERT	0.807	0.777	0.576	0.662	0.921	**0.781**
XLM-R	0.883	0.883	0.675	0.692	**0.981**	**0.865**
InfoXLM	**0.921**	**1.093**	0.698	0.702	0.900	0.662
MuRIL	0.778	0.728	0.575	0.604	**0.990**	0.704
BLOOM	**0.957**	**0.903**	**0.731**	**0.742**	0.966	0.707
mT5	**0.961**	0.790	**0.773**	**0.767**	0.952	**0.757**
mGPT	0.950	**1.040**	**0.728**	**0.736**	0.946	0.723
XGLM	0.954	0.972	0.724	0.730	0.956	0.715

Universal multi-lingual models: show greater resilience to perturbations in at least four languages

BERT-specific models: display a more significant accuracy drop across all the Indic languages

Perturbation Results: layer-wise robustness analysis

Probing tasks vs. Indic languages

	hi	kn	ml	mr	te	ur
SentLen	1, 2, 3	4, 3, 2	3, 4, 1	1, 3, 2	1, 2	3, 1, 2
TreeDepth	7, 5, 6	5, 4, 9	4, 5, 3	4, 8, 9	10, 11, 12	3, 2, 5
VerbGen	10, 8, 5	11, 4, 12	-	11, 10, 12	6, 8, 9	3, 8, 9
VerbNum	Equal	3, 4, 5	-	2, 4, 5	10, 12	5, 4, 3
VerbPer	Equal	Equal	-	3, 4, 5	5, 11, 12	4, 5
SubjNum	N/A	Equal	9, 8, 7	7, 10, 9	10, 11	1, 2
ObjNum	11, 12	4, 3, 9	Equal	11, 10, 12	11, 12	10, 9, 11

Surface: early layers are impacted across Indic languages

Syntactic: middle layers are affected for all languages except **te**

Semantic: for **ur**-early to middle, while middle to later layers for other languages

Conclusion

- For **En: probing analysis** reveals that almost **all multilingual models** demonstrate **consistent encoding performance** and **similar language hierarchy**
- For **Indic languages**, **Indic-specific multilingual models** capture better **language hierarchy** while **universal models** show mixed results.
- Intriguingly, **universal models** broadly exhibit **better robustness** compared to **Indic-specific models**